

3-Mark Questions & Answers — Data Structures Module 5: Graphs

Q1. Compare BFS and DFS graph traversal algorithms.

Feature	Breadth-First Search (BFS)	Depth-First Search (DFS)
Data Structure	FIFO Queue	Call Stack / Recursion
Strategy	Level-by-level exploration	Branch-by-branch deep dive
Shortest Path	Finds shortest path in unweighted graphs	Explores deep paths & cycles
Complexity	$O(V + E)$	$O(V + E)$

Q2. Explain Kahn's algorithm for Topological Sorting.

1. Compute in-degree for all vertices.
2. Push all vertices with in-degree 0 into a Queue.
3. Dequeue vertex `u`, add to result, and decrement in-degree of all neighbors `v`.
4. If in-degree of `v` becomes 0, enqueue `v`. Repeat until queue is empty.